Fiscal Year:	FY 2023	**Task Last Updated:**	FY 09/27/2022
PI Name:	Zwart, Sara Ph.D.		
Project Title:	Exploring Endothelial Function Risk Factors and Optic Disc Edema Changes During Strict 6º Head-Down Tilt Bed Rest		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77058-3607	**Congressional District:**	36
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2020 HERO 80JSC019N0001-HFBP, OMNIBUS3 Crew Health: Human Factors and Behavioral Performance-Appendix E; Omnibus3-Appendix F
Start Date:	02/23/2021	**End Date:**	05/30/2022
No. of Post Docs:	0	**No. of PhD Degrees:**	0
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No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 5/30/2022 per C. Ribeiro/JSC. (Ed., 12/14/21)		
Key Personnel Changes/Previous PI:	December 2021 report: No changes.		
COI Name (Institution):	Macias, Brandon Ph.D. (NASA Johnson Space Center) Melin, M. Mark M.D. (University of Minnesota) Smith, Scott Ph.D. (NASA Johnson Space Center) Young, Millennia Ph.D. (NASA Johnson Space Center) Bershad, Eric (Baylor College of Medicine)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Approximately 20% of astronauts on International Space Station (ISS) missions experience ophthalmic pathologies including optic disc edema, part of what is characterized as Spaceflight Associated Neuro-ocular Syndrome (SANS). While the cause of SANS is unknown, there are likely multiple contributing factors, including genetics, that may affect the response to spaceflight in affected individuals. B-vitamin status and the presence of specific one-carbon pathway single nucleotide polymorphism (SNP) alleles predicted the incidence of SANS pathologies. When looking at the individual SNPs, the G allele of methionine synthase reductase (MTRR, rs1801394) A66G and the C allele of serine hydroxymethyltransferase-1 (SHMT1, rs1979277) C1420T were associated with higher incidence of SANS compared to those with the A or T alleles. Supporting data from a recently published 30-d bed rest study demonstrated the same alleles were related to the magnitude of optic disc edema estimated as the change in total retina thickness (? TRT). We propose here to study the same genetic variants in another bed rest study that has already completed: Artificial Gravity Bed Rest – European Space Agency (AGBRESA). Some of those subjects also developed optic disc edema. Furthermore, we will test available urine samples from prior bed rest studies (VaPER (VIIP and Psychological :envihab Research) and AGBRESA) for markers of glycocalyx degradation to expand our understanding of this problem. For information on AGBRESA and the :envihab facility: https://
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Knowledge gained from this experiment will contribute to a better understanding of endothelial function and its possible relationship with optic disc edema, a condition that affects some individuals in the general population.
Task Progress:	Approximately 20% of astronauts on International Space Station (ISS) missions experience ophthalmic pathologies, including optic disc edema, part of what is characterized as Spaceflight Associated Neuro-ocular Syndrome (SANS). B-vitamin status and the presence of specific one-carbon pathway single nucleotide polymorphism (SNP) alleles predicted the incidence of SANS pathologies in astronauts and in previous head-down tilt (HDT) bed rest subjects. In this study, 24 subjects participated in a 60-day HDT bed rest study. Blood samples were collected up to 1 year after the study for genetic analysis. The statistical model used from a previous bed rest study to assess the relationship between total retinal thickness and one carbon pathway genetics was applied to data from the present study. The statistical model produced similar results, in that subjects with 3-4 risk alleles (G and C alleles from the MTRR A66G and SHMT1 C1420T SNPs) had a greater change in total retinal thickness during and after bed rest compared to subjects with 0-2 risk alleles. A second aim of this study was to investigate whether there is any evidence that the glycocalyx, the 0.5 (capillaries) to 5 (conductance arteries) micron thick coating which lines the entirety of the vascular lumen and has significant effects on endothelial function, is compromised during bed rest. Vascular distension, stasis, and altered shear forces, all of which are associated with headward fluid shifts, may also contribute to glycocalyx dysfunction and shedding, resulting in endothelial dysfunction and increased vascular permeability and tissue edema, potentially contributing to optic nerve and optic disc edema in affected individuals. Spot urine samples from 13 female subjects from 2 bed rest studies where SANS was evident in some individuals were assessed for markers of glycocalyx degradation. Neither total glycosaminoglycans nor hyaluronic acid were different in these spot urine samples from subjects during bed rest. This study provides evidence from a second bed rest study demonstrating the association of the MTRR and SHMT1 genetic variants with ocular pathologies during bed rest. These data reaffirm that genetics and B vitamin status predispose some individuals to develop SANS, and offer opportunities for countermeasure testing, where supplemental nutraceuticals have the potential to override the genetic influences on biochemistry, physiology, and pathophysiology.
Bibliography Type:	Description: (Last Updated: 07/25/2025)
Articles in Peer-reviewed Journals	Smith SM, Zwart SR. "Nutrition as fuel for human spaceflight." Physiology (Bethesda). 2021 Sep 1;36(5):324-30. Review. https://doi.org/10.1152/physiol.00011.2021 ; PMID: 34431417 , Sep-2021